

- 4 Coherent light of wavelength 590 nm is incident normally on a double slit, as shown in Fig. 4.1.

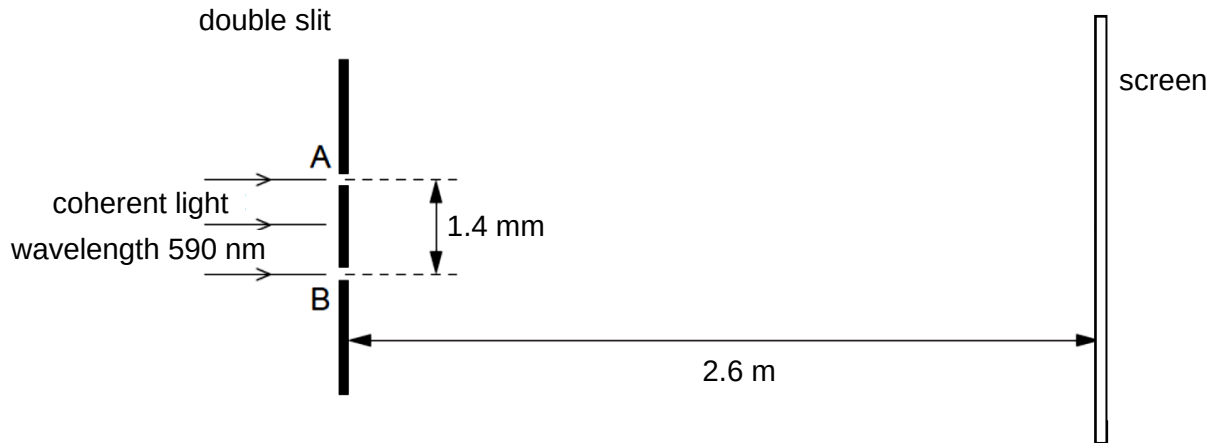


Fig. 4.1

The separation of the slits A and B is 1.4 mm and the width of each slit is 0.15mm. A screen is placed parallel to the slits at a distance 2.6 m away.

(a) One of the slits is covered.

Calculate the width of the central maximum formed on the screen by diffraction through the uncovered slit.

$$\begin{aligned}\sin \theta_{\min} &= \frac{\lambda}{b} \\ &= \frac{5.90 \times 10^{-7}}{0.15 \times 10^{-3}} \\ &= 0.00393 \quad [\text{M1}] \\ \theta_{\min} &= 0.225^\circ\end{aligned}$$

$$\begin{aligned}\tan 0.225^\circ &= \frac{1/2 w}{2.6} \quad [\text{M1}] \\ w &= 2(2.6 \times \tan 0.225^\circ) \\ &= 0.0204 \text{ m} \quad [\text{A1}]\end{aligned}$$

width = cm [3]

(b) Now, both slits are uncovered.

(i) State Rayleigh's criterion.

Rayleigh's criterion says that the images of two-point sources can just be distinguished/ resolved [B1] from each other if the central maximum of one diffraction pattern falls on the first minimum of the other one. [B1] [2]

(ii) Use Rayleigh's criterion to explain whether the diffraction pattern produced by the two slits are seen as on the screen as separate.

Since the separation of the double slits is 1.4 mm, the central maxima of the two diffraction patterns will be 1.4 mm apart. [B1]

From (a) the first minimum of each pattern is at a distance of 10.2 mm away from the central maximum of the pattern. [B1]

Hence, the maximum of the first diffraction will fall within the 1st minimum of other diffraction pattern. The two patterns are not distinguished/ resolved and will not be seen as separate. [B1] [3]

[Total: 8]

Fig. 5.1 shows the arrangement of the apparatus used in the measurement of the magnetic flux.